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The Influence of Sport Representation and Attitudes Toward Strength Training on Neuromuscular Performance Profiles in University Athletes: Part I Female Athletes

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Abstract

McClean, ZJ, da Silva Torres, R, Herzog, W, Pasanen, K, Lun, V, Nimphius, S, and Jordan, MJ. The influence of sport representation and attitudes toward strength training on neuromuscular performance profiles in university athletes: part I female athletes. *J Strength Cond Res* XX(X): 000–000, 2025—Neuromuscular performance is often assessed across strength, power, and plyometric tasks in athlete populations with an increasing emphasis on biomechanical testing that yields multifaceted data sets with several outcome measures per task. This leads to specific challenges in determining relevant neuromuscular performance benchmarks especially when considering contextual factors related to the sport environment. Thus, characterizing athlete neuromuscular performance subpopulations (or profiles) while accounting for the influence of contextual factors such as those related to the sport environment, and psychosocial factors such as the athlete's attitudes toward strength training could provide a framework for more specific neuromuscular benchmarks. Therefore, neuromuscular performance profiles were explored in female university athletes while accounting for sport influences and attitudes toward strength training. Female university athletes ($n = 192$) from 5 sports completed a comprehensive lower body neuromuscular performance testing battery and surveys assessing attitudes toward strength training. Clustering was done to define neuromuscular performance profiles and Fisher's Exact Tests were used to examine differences in attitudes toward strength training across clusters. A high-strength/high-power/high-stiffness cluster, low-strength/low-power/low-stiffness cluster, and low-strength/low-power/high-stiffness cluster were identified with no difference in sport representation but differences in attitudes toward strength training ($p < 0.047$; alpha = 0.05). These findings may help to quantify the characteristics of different female university athlete neuromuscular performance profiles while accounting for potential sport and psychosocial influences. One potential application of these results is to inform neuromuscular performance benchmarks that can be used in conjunction with preseason evaluations or in the context of restoring performance readiness after injury.

Key Words: athlete profiling, countermovement jump testing, sport injury, sport performance, unsupervised machine learning

Introduction

Lower body neuromuscular performance is often assessed across tests of maximal strength, power, and plyometric capacity in competitive athlete populations, including university and college athletes (1,30). There is also an increasing emphasis on biomechanical testing with force plates to quantify neuromuscular performance related to coupled eccentric–concentric movements known as stretch-shortening cycles (3,21). This approach yields multifaceted data sets with a high number of outcome measures that can be used to inform training (11,29), monitor neuromuscular fatigue (11,29), and to quantify return-to-sport readiness and performance readiness after injury (11). Furthermore, a comprehensive neuromuscular performance assessment that includes biomechanical testing is particularly useful after severe injuries such as anterior cruciate ligament (ACL) rupture where prolonged performance impairments are noted (2,44) across strength (16,33,35,47) and jumping tasks (11,13,15,27,29,32,46). However, the abundance of outcome measures obtained from this comprehensive testing methodology may pose challenges for defining relevant benchmarks across tasks, and an overreliance on interpreting single outcome measures in isolation may be problematic (4,11).

An alternative approach is to define relevant athlete subpopulations (or profiles) using the combined weighted expression of their neuromuscular performance characteristics (4). The notion of an athlete profile has been discussed in sport science literature, but there is limited discussion on the best methodologic approach (45). Unsupervised machine learning (ML) is a viable method for defining athlete profiles because it accounts for the weighted pattern in the expression of multiple performance variables within a complex data set similar to that obtained from neuromuscular performance assessments. For instance, Jauhiainen et al. (17) used this approach to define biomechanical movement profiles of runners, while Wilk

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et al. (50) applied unsupervised ML to develop anthropometrical and physiologic profiles of swimmers for performance prediction, and Martin et al. (26) established ACL surgical profiles that considered graft type, damage to secondary tissues, and patient perception of knee function to understand patterns influencing postsurgical recovery. The application of unsupervised ML seems promising to explore the existence of neuromuscular performance profile subpopulations in female university athletes. In addition to being used as a framework to guide the interpretation of preseason performance testing, this approach may also be useful for defining profile-specific performance-readiness benchmark criteria after severe injuries such as ACL rupture.

In addition, examining athlete neuromuscular performance profiles in the context of the athlete's sport environment (e.g., sport specificity) and attitudes toward strength training, with the latter being particularly relevant for female, woman, and girl athletes, may provide valuable insights into potential barriers for engaging in heavy strength training among female university athletes (36,49). Furthermore, assessments that quantify psychosocial contextual factors may also include an analysis of the athlete's perception of their overall musculoskeletal function perceived limb function using the Sport Fitness Index (SFI), which was previously developed for injury risk assessment (51) and applied in conjunction with neuromuscular performance assessments in a university athlete population (27).

Therefore, the purpose of this study was to (a) to apply an unsupervised ML cluster analysis to a comprehensive neuromuscular performance assessment incorporating biomechanical assessments of lower body maximal strength, power, and plyometric capacities in a cohort of healthy, uninjured female university athletes and (b) compare psychosocial and contextual factors between neuromuscular performance profiles including sport representation, attitudes toward strength training, and perceived limb function assessed with the SFI.

Methods

Experimental Approach to the Problem

The neuromuscular performance assessment was performed as a part of a multiyear longitudinal university athlete monitoring program for 2 consecutive years. Healthy, uninjured female athlete subjects were familiarized with neuromuscular performance assessments that included countermovement jump (CMJ) testing with and without additional external load, single leg CMJ testing, single leg repeat hop testing, and multijoint leg extensor eccentric, isometric, and concentric maximal strength testing on a robotic leg press. Before baseline testing, subjects completed a standardized warm-up (27) that included self-selected static and dynamic stretching of the lower body musculature, 10 minutes on a cycle ergometer at a self-selected moderate intensity, 4 sets of 10 seconds cycling sprints interspersed by 50 seconds of active rest, and a task-specific warm-up that included single leg and double leg CMJs, and double leg repeat hopping. Subjects completed surveys after the neuromuscular performance assessment that included demographic information (i.e., sex, age) and subject demographics (Table 1). All subjects identified as women. In addition, subjects completed the SFI questionnaire (51), and a survey evaluating athlete attitudes toward strength training and injury prevention, which was adapted from the Players' Attitudes Toward Injuries, Training, and Injury Prevention questionnaire from Møller et al. (34) and Pasanen (37).

Subjects

Healthy female university athletes from 5 team sports (soccer, basketball, volleyball, rugby, and field hockey) were recruited using a cluster sampling approach and provided their written informed consent. Baseline testing was conducted for 2 consecutive years with 150 unique athletes participating in the study. Forty-two university athletes who competed in U Sports during both years/seasons were, therefore, included in baseline testing twice, resulting in a total sample size of 192 athletes. Subjects were included in the study if they were (a) in the preseason phase of the off-season and preparing for competition in the upcoming university sport season, (b) cleared by an athletic therapist to perform maximal strength testing and vertical jump testing, and (c) had not sustained an ACL injury in the past 24 months. Subjects were excluded from the study if they had any musculoskeletal conditions, such as traumatic knee injuries, which limited their ability to perform lower limb maximal strength or vertical jump testing. The protocol for this study was approved by the research institution's Conjoint Health Research Ethics Board (REB15-1094) and conducted in accordance with the Declaration of Helsinki (without registration).

Procedures

Countermovement Jump Testing. For all CMJ trials, subjects stood on a dual force plate system (Accupower, Advanced Measurement Technology Instruments, Boston, MA, sampling frequency = 1000 Hz) and performed a series of jumps that included bilateral CMJ testing (CMJ, repetitions = 5), single leg CMJ testing (single leg body weight countermovement jump [CMJ_{SL}], repetitions = 3/leg), and double leg CMJ testing with an extra external load equal to 60% of the subject's body weight (CMJ_{loaded}, repetitions = 3). Subjects were instructed to descend rapidly to a self-selected depth and ascend with a maximum effort and to maximize their jump height (5,20,22). The start of each CMJ repetition was indicated by a strong verbal cue by the researcher, and each CMJ repetition was followed by a 5-second period where the subject stood still on the force plate system. Countermovement jump and single leg bodyweight countermovement jump trials were conducted with the subjects' hands on their hips, and for CMJ_{loaded} trials, subjects held the high handles of a hexagonal bar equal to 60% of their body weight.

Vertical ground reaction forces (Fz) for each force plate were measured using the Noraxon Myoresearch interface (version 3.20, Scottsdale, AZ). Raw data were analyzed using a custom-built computer program (MATLAB R2022a, Mathworks, Natwick, MA), which has been described previously (20). In short, body mass of the subjects was calculated as the total Fz from the left and right force plates during a 2-second period where the subject stood still (20,21). Countermovement jump phases were delineated based on the velocity of the body's center of mass (BCM) (calculated as the integral of the acceleration signal derived from the summed bilateral Fz signal and verifying that the BCM velocity was zero at the onset of each CMJ) (20). The CMJ eccentric deceleration phase was defined as the time from the BCM at maximal negative velocity to the BCM at zero velocity (20). The CMJ concentric phase was defined as the time from the BCM at zero velocity to the point of CMJ takeoff (20). Jump height was calculated using takeoff velocity (Jump height = takeoff velocity^2/(2 g)) and system mechanical power was calculated continuously as the product of Fz and BCM vertical velocity. The maximum of the absolute system mechanical power was calculated for the concentric and eccentric

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Table 1

Neuromuscular characteristics of 3 female university athlete clusters measured across 5 testing batteries that evaluated maximal strength, power, and plyometric function.*†

Testing protocol	Outcome measure	Cluster centroid value (mean ± *SD*)		
		$Cluster_{WEAK-FAST}$	$Cluster_{WEAK-SLOW}$	$Cluster_{STRONG-FAST}$
	Age (y)	20.0 ± 1.5	20.1 ± 2.1	19.9 ± 1.4
	Body mass (kg)	65.9 ± 10.1	71.3 ± 11.6	68.7 ± 9.4
Body weight CMJ	Jump height (cm)	23.8 ± 4.0	23.3 ± 4.7	29.8 ± 4.4
	Peak propulsive power ($W \cdot kg^{-1}$)	41.9 ± 4.7	39.5 ± 5.3	47.6 ± 5.1
	Rate of power development ($W \cdot kg^{-1} \cdot s^{-1}$)	224.2 ± 58.8	166.2 ± 40.4	247.1 ± 45.4
	Net eccentric deceleration phase impulse ($Ns \cdot kg^{-1}$)	0.97 ± 0.16	1.05 ± 0.20	1.23 ± 0.18
	Peak braking power ($W \cdot kg^{-1}$)	12.1 ± 2.7	12.8 ± 3.4	17.5 ± 4.0
	Concentric phase duration (s)	0.26 ± 0.03	0.31 ± 0.03	0.26 ± 0.03
	Eccentric deceleration phase duration (s)	0.17 ± 0.03	0.20 ± 0.03	0.15 ± 0.02
	Contraction time (s)	0.81 ± 0.10	0.94 ± 0.10	0.82 ± 0.11
	Lower limb eccentric deceleration stiffness ($[N \cdot m^{-1}] \cdot kg^{-1}$)	111.9 ± 41.1	73.5 ± 19.4	118.0 ± 34.1
	Force at zero velocity ($N \cdot kg^{-1}$)	21.0 ± 2.1	19.8 ± 1.9	23.4 ± 2.2
	Vmax − TOV percentage difference (%)	7.6 ± 1.6	8.1 ± 1.7	6.4 ± 1.5
CMJ plus additional external load (60% body weight)	Jump height (cm)	11.5 ± 2.9	10.9 ± 3.1	16.2 ± 3.3
	Peak propulsive power ($W \cdot kg^{-1}$)	42.2 ± 5.0	39.9 ± 5.9	49.3 ± 5.8
	Rate of power development ($W \cdot kg^{-1} \cdot s^{-1}$)	155.9 ± 28.7	109.0 ± 25.8	172.4 ± 34.1
	Net eccentric deceleration phase impulse ($Ns \cdot kg^{-1}$)	1.24 ± 0.25	1.29 ± 0.22	1.60 ± 0.25
	Peak braking power ($W \cdot kg^{-1}$)	13.8 ± 3.2	14.0 ± 2.8	18.8 ± 4.0
	Concentric phase duration (s)	0.37 ± 0.04	0.47 ± 0.06	0.38 ± 0.04
	Eccentric deceleration phase duration (s)	0.24 ± 0.04	0.32 ± 0.05	0.24 ± 0.04
	Contraction time (s)	1.03 ± 0.12	1.24 ± 0.16	1.03 ± 0.12
	Lower limb eccentric deceleration stiffness ($[N \cdot m^{-1}] \cdot kg^{-1}$)	155.9 ± 28.7	109.0 ± 25.8	172.4 ± 34.1
	Force at zero velocity ($N \cdot kg^{-1}$)	24.7 ± 1.7	22.9 ± 1.4	26.6 ± 2.13
	Vmax − TOV percentage difference (%)	16.3 ± 4.0	17.0 ± 3.8	12.4 ± 3.0
Single leg, body weight CMJ	Jump height (cm)	11.7 ± 2.6	10.5 ± 2.6	14.8 ± 2.6
	Peak propulsive power ($W \cdot kg^{-1}$)	26.2 ± 2.7	24.2 ± 3.2	30.0 ± 3.5
	Rate of power development ($W \cdot kg^{-1} \cdot s^{-1}$)	140.2 ± 35.9	95.5 ± 21.2	144.3 ± 30.4
	Net eccentric deceleration phase impulse ($Ns \cdot kg^{-1}$)	0.67 ± 0.12	0.73 ± 0.13	0.88 ± 0.13
	Peak braking power ($W \cdot kg^{-1}$)	7.7 ± 1.6	8.2 ± 2.0	11.0 ± 2.0
	Concentric phase duration (s)	0.31 ± 0.04	0.38 ± 0.04	0.32 ± 0.04
	Eccentric deceleration phase duration (s)	0.17 ± 0.03	0.23 ± 0.04	0.18 ± 0.03
	Contraction time (s)	0.86 ± 0.11	1.01 ± 0.10	0.90 ± 0.12
	Lower limb eccentric deceleration stiffness ($[N \cdot m^{-1}] \cdot kg^{-1}$)	146.7 ± 75.5	73.1 ± 30.0	108.3 ± 47.4
	Force at zero velocity ($N \cdot kg^{-1}$)	17.5 ± 1.7	15.8 ± 1.0	18.1 ± 1.5
	Vmax − TOV percentage difference (%)	15.6 ± 4.4	18.2 ± 5.3	13.2 ± 3.1
Single leg repeated hop	Jump height (cm)	7.4 ± 2.4	8.8 ± 2.8	11.7 ± 2.6
	Contact time (s)	0.27 ± 0.04	0.33 ± 0.05	0.31 ± 0.05
	Peak ground reaction force ($N \cdot kg^{-1}$)	34.5 ± 3.8	31.9 ± 4.0	34.2 ± 3.9
	Lower limb eccentric deceleration stiffness ($[N \cdot m^{-1}] \cdot kg^{-1}$)	324.5 ± 78.5	232.3 ± 83.2	231.3 ± 90.6
Maximal multijoint leg press strength	Concentric MVC ($N \cdot kg^{-1}$)	20.1 ± 2.3	20.3 ± 3.3	22.3 ± 3.7
	Eccentric MVC ($N \cdot kg^{-1}$)	24.0 ± 4.6	24.7 ± 4.6	25.9 ± 4.6

*CMJ = countermovement jump; TOV = velocity at takeoff.

†Cluster 1 ($Cluster_{WEAK-FAST}$) displayed low muscle strength and fast contraction speeds, Cluster 2 ($Cluster_{WEAK-SLOW}$) displayed low muscle strength and slow contraction speeds, Cluster 3 ($Cluster_{STRONG-FAST}$) displayed high muscle strength and fast contraction speeds.

CMJ phases. Lower limb eccentric deceleration stiffness (LL_{STIFF}) was calculated as the change in Fz divided by the change in the downward displacement of the body center of mass during the eccentric deceleration phase ($N \cdot m^{-1}$), which was then body mass normalized for the analysis. Force at zero velocity was defined as the Fz value at the onset of the concentric phase. The percentage difference between the maximal velocity obtained during the concentric phase (Vmax) and the velocity at takeoff (TOV) was used to calculate the Vmax − TOV difference.

Single Leg Repeat Hop Testing. Subjects performed 20 continuous repetitions of single leg repeat hop testing (RH_{SL}) on each leg on the same force plate system. Subjects were cued to maximize the hop height while minimizing ground contact time and keeping their hands on their hips. The resulting force–time curves were low-pass filtered using an inverse second order Butterworth filter (frequency = 50 Hz) to minimize the ground reaction force landing artifact. The midpoint of each hop's flight phase was assumed to represent maximal hop height where the velocity of the BCM is equal to zero. The hop height was then used to infer the subsequent hop's landing velocity so that discrete hop movement phases could be calculated using the BCM velocity. The average hop performance over the middle 16 hops on each leg was used for analysis. Then, the average between the calculated

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left leg value and the calculated right leg value was used for analysis.

Single Leg Isokinetic Leg Extension Testing. Subjects underwent single leg isokinetic concentric and eccentric maximal strength testing on a multijoint robotic servomotor leg press device (Fv Seger, Treadmetrix, Park City, UT; sampling frequency = 200 Hz). Subjects positioned themselves on the semirecumbent leg press chair with their feet set on the leg press platform in a standardized position. The subject's leg extension range of motion was set for knee joint angles between 10 and 100 degrees of flexion. Before testing, a task-specific warm-up protocol was completed to familiarize the subjects. The warm-up included 4 concentric–eccentric isokinetic repetitions at 0.2 $m \cdot s^{-1}$ conducted at progressively increasing intensities (50, 70, 90% of the subject's perceived maximal effort). In addition, 3 single leg isometric warm-up contractions at a 60-degree knee flexion angle were completed for each leg.

After a 3–5-minute rest period, and using a sled velocity (0.2 $m \cdot s^{-1}$) (42), isokinetic testing was performed for concentric and eccentric leg press actions. Three repetitions were conducted with a 0.5-second pause between the concentric and eccentric movement phases. During isokinetic tests, the cue "push as hard as possible throughout the duration of the repetition" and strong verbal cues of "push, push!" during concentric contractions, and "resist, resist!" during eccentric contractions were used. The peak concentric (CON_{MVC}) and eccentric (ECC_{MVC}) force values across all repetitions were obtained to reflect maximal concentric and eccentric muscle strength capacities.

Data Analysis. Neuromuscular performance values laying more than ±3 *SD*s from the mean were removed. Approximately 0.15% of all concentric strength, eccentric strength, CMJ, CMJ_{SL}, CMJ_{loaded}, and RH_{SL} values were removed as outliers. For CMJ, CMJ_{SL}, CMJ_{loaded}, and RH_{SL} testing, performance metrics were averaged across all trial values included for analysis (9). For maximal concentric and eccentric muscle strength, the maximal force production value obtained during left leg and right leg testing was used for analysis (27,28).

Questionnaires. The original 10-question SFI (51) was completed by all subjects. This survey assesses the influence of multiple biopsychosocial constructs on an athlete's perception of their overall musculoskeletal function and is retrospectively and prospectively predictive of sport injury (51) and lower limb biomechanics (27). To compute a final SFI score, each question was numerically coded and scores were summed for a final score between 0 and 100, with higher scores reflecting better perception of limb function (51).

The Attitudes Toward Strength Training and Injury Prevention questionnaire was adapted from Møller et al. (34) and Pasanen (37) to evaluate attitudes toward strength training and injury prevention. This questionnaire consists of 22 questions related to athlete attitudes toward sport injury risk, injury prevention training, and strength training outcomes. Each question was answered using a 7-point Likert scale with options ranging from "Strongly agree" to "Strongly disagree." To isolate subjects with more extreme responses, we categorized the 7-point Likert scale into 3 broader groups: (a) strongly agree and agree responses were grouped to reflect high agreement with the statement; (b) slightly agree, neither, and slightly disagree responses were grouped to reflect a moderate stance toward the statement; and (c) strongly disagree and disagree responses were grouped to reflect high disagreement with the statement.

Data Processing. Before analysis, a matrix of biomechanical features was created (192 subjects × 39 features) of which a total of 0.8% of the observations were missing. In all cases, missing data were because of subjects declining to do either the CMJ_{loaded}, RH_{SL}, or maximal strength testing protocols. All relevant features were normalized to the athlete's body mass, then features were scaled according to the interquartile range using the RobustScaler function from Scikit-learn (14) (v. 0.20.4). Missing data were imputed based on the mean value of the 5 nearest neighbors in the data set using the KNNImputer function (48) from Scikit-learn.

Injury Surveillance. Injury surveillance was performed in collaboration with the university's athletic therapy department to identify athletes sustaining noncontact ACL injuries. Injuries were recorded on a weekly basis by the athletic therapy department using an electronic medical record system. The athletic therapy staff identified individual athletes who went on to sustain noncontact ACL injuries within the season after that year's baseline testing. Noncontact ACL injury was defined following previous recommendations (38) and included scenarios where there was no direct contact to the athlete (i.e., sudden onset) and scenarios where the athlete was contacted by an athlete or object at the time of injury, but the contact did not occur directly to the injured area. Athlete exposure to training load, game load, or high-risk events for injury was not recorded.

Statistical Analyses

Descriptive statistics are presented for the group means and standard deviation. Statistical analyses were conducted using The Python Language (v. 3.14). Measures of cluster quality (i.e., Silhouette score, Davies Bouldin Score, and Calinski Harabasz Score) were calculated for each combination of dimensionality reduction algorithm (i.e., Principal Component Analysis, t-distributed stochastic neighbor embedding, and Uniform Manifold Approximation and Projection for Dimension Reduction [UMAP]) and clustering algorithm (i.e., K-means, hierarchical clustering analysis, Density-based Spatial Clustering of Applications with Noise, and Hierarchical Density-based Spatial Clustering of Applications with Noise) to determine the method that produced the most distinct clustering outcome. The combination of UMAP (31) (neighbors = 10, components = 15) and K-means clustering (39) (clusters = 3) produced the highest silhouette score, and the selected number of clusters was confirmed using the K-means clustering "elbow method" (10).

To explain the relative contribution of each feature in the determination of cluster assignment (i.e., model explainability), the recently recommended neutralization–propagation (NEON) procedure (23) was used. The top 20 most important features for each cluster's centroid (i.e., the average value for each cluster) were computed. Linear regression modeling was used to (a) examine the effect of athlete age, experience level (i.e., number of years competing in elite sport), body mass, and SFI score on cluster assignment and (b) examine the effect of athlete experience level on Attitudes Toward Non-Sport Specific Training. Finally, Fisher's Exact Tests were conducted to examine differences in Attitudes Toward Non-Sport Specific Training across athlete clusters. For all applicable statistics, an alpha level of 0.05 was used as the threshold to indicate significance.

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Results

Three neuromuscular performance clusters were identified using the K-means clustering approach (Silhouette score = 0.51, Davies Bouldin Score = 0.74, Calinski Harabasz Score = 287). A 2-dimensional representation of the cluster output is presented in Figure 1, and the characteristics of each cluster are given in Table 1.

Cluster 1: Low Strength and Power, Fast Jump Contraction Time ($Cluster_{WEAK\text{-}FAST}$)

$Cluster_{WEAK\text{-}FAST}$ (n = 47; 24.5% of total sample) was made up of soccer (38.3% of the cluster), rugby (27.7% of the cluster), field hockey (17.0% of the cluster), basketball (12.8% of the cluster), and volleyball (4.3% of the cluster) athletes. This was equivalent to 40.0% of the total soccer population, 21.3% of the total rugby population, 21.0% of the total field hockey population, 26.1% of the total basketball population, and 8.1% of the total volleyball population. The top 20 features describing the biomechanical characteristics of this cluster are shown in Figure 2. $Cluster_{WEAK\text{-}FAST}$ exhibited the highest high lower limb eccentric deceleration stiffness (CMJ_{loaded}: 81.8 ± 26.0 $[N{\cdot}cm^{-1}]{\cdot}kg^{-1}$; CMJ_{SL}: 146.7 ± 75.5 $[N{\cdot}cm^{-1}]{\cdot}kg^{-1}$; RH_{SL}: 324.5 ± 78.5 $[N{\cdot}cm^{-1}]{\cdot}kg^{-1}$), the shortest contraction times (CMJ_{loaded}: 1.03 ± 0.12 seconds; CMJ_{SL}: 0.86 ± 0.11 seconds; RH_{SL}: 0.27 ± 0.04 seconds), and below average jump performance (i.e., jump height; CMJ: 23.8 ± 4.0 cm; CMJ_{loaded}: 11.5 ± 2.9 cm; CMJ_{SL}: 11.7 ± 2.6 cm; RH_{SL}: 7.4 ± 2.4 cm). In addition, $Cluster_{WEAK\text{-}FAST}$ demonstrated the lowest maximal concentric strength (CON_{MVC}: 20.1 ± 2.3 $N{\cdot}kg^{-1}$) compared with the other clusters.

Cluster 2: Low Strength and Power, Slow Jump Contraction Time ($Cluster_{WEAK\text{-}SLOW}$)

$Cluster_{WEAK\text{-}SLOW}$ (n = 93; 48.4% of total sample) was made up of rugby (37.6% of the cluster), field hockey (32.3% of the cluster), soccer (17.2% of the cluster), basketball (9.7% of the cluster), and volleyball (3.2% of the cluster) athletes. This was equivalent to 57.3% of the rugby population, 79.0% of the field hockey population, 35.5% of the soccer population, 39.1% of the basketball population, and 11.9% of the volleyball population. Athletes in $Cluster_{WEAK\text{-}SLOW}$ (Figure 3) demonstrated the longest contraction times in both the eccentric deceleration (CMJ: 0.2 ± 0.03 seconds; CMJ_{loaded}: 0.32 ± 0.05 seconds; CMJ_{SL}: 0.28 ± 0.04 seconds) and concentric (CMJ: 0.31 ± 0.03 seconds; CMJ_{loaded}: 0.47 ± 0.06 seconds; CMJ_{SL}: 0.38 ± 0.04 seconds) phases, along with below average jump performance (jump height: CMJ = 23.3 ± 4.7 cm, CMJ_{loaded} = 10.9 ± 3.1 cm, CMJ_{SL} = 10.5 ± 2.6 cm).

Cluster 3: High Strength and Power, Fast Jump Contraction Time ($Cluster_{STRONG\text{-}FAST}$)

$Cluster_{STRONG\text{-}FAST}$ (n = 52; 27.1% of total sample) was made up of volleyball (38.5% of the cluster), rugby (25.0% of the cluster), soccer (21.2% of the cluster), and basketball (15.4% of the cluster) athletes. This was equivalent to 80.0% of the volleyball population, 21.3% of the rugby population, 24.5% of the soccer population, 34.7% of the field hockey population, and 0.0% of the field hockey population. Biomechanical features of $Cluster_{STRONG\text{-}FAST}$ (Figure 4) described a below average jump

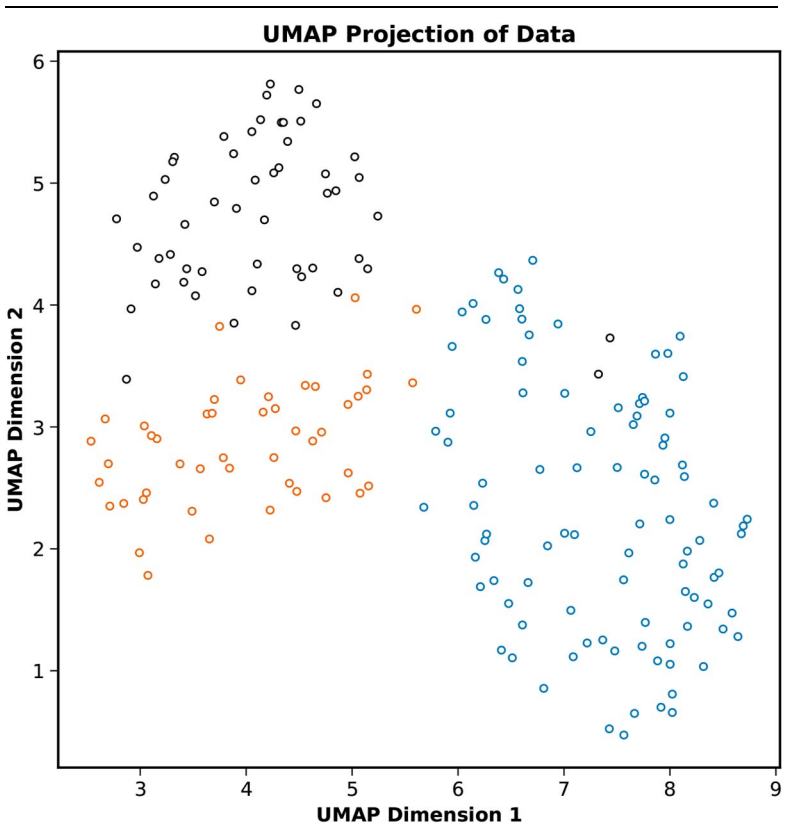


Figure 1. Two-dimensional visual representation of clustering assignments using uniform manifold approximation and projection for dimension reduction (UMAP). Colors are representative of different athlete clusters.

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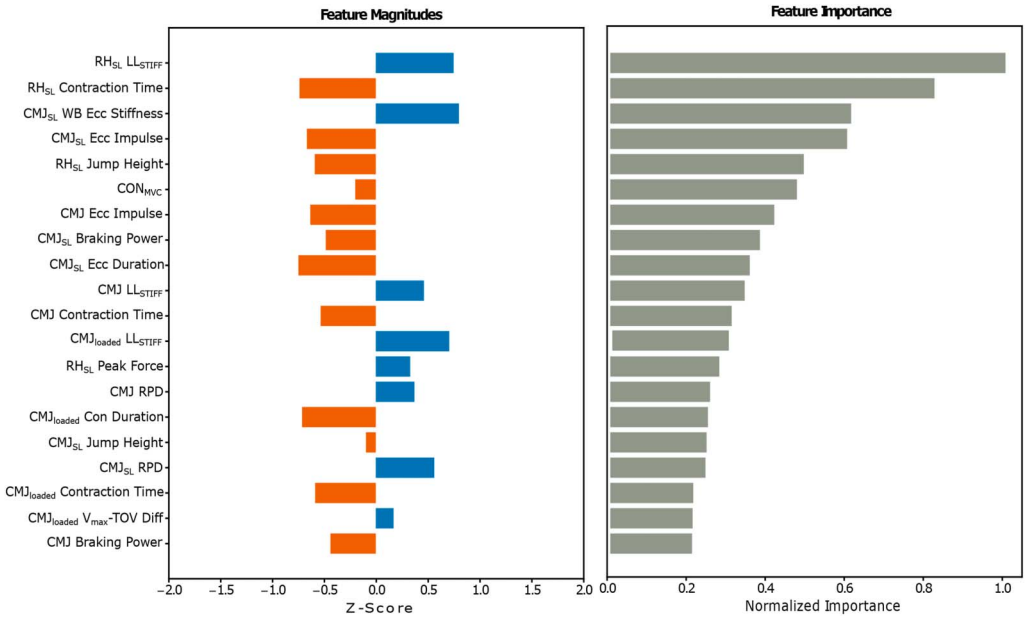


Figure 2. The average feature importance for the top 20 most important features in assigning athletes to Cluster$_{WEAK\text{-}FAST}$. CMJ = body weight countermovement jump; CMJ_{SL} = single leg body weight countermovement jump; CMJ_{loaded} = countermovement jump plus additional external load (60% body weight); RH_{SL} = single leg repeated hop; CON_{MVC} = concentric multijoint maximal voluntary contraction; ECC_{MVC} = eccentric multijoint maximal voluntary contraction; Con = concentric; Ecc = eccentric; LL_{STIFF} = lower body eccentric deceleration stiffness; RPD = rate of propulsive power development; Vmax − TOV Diff = percentage difference between the maximal velocity obtained during the concentric phase (Vmax) and the velocity at takeoff (TOV); concentric and eccentric impulses are calculated as the net concentric and eccentric deceleration impulse.

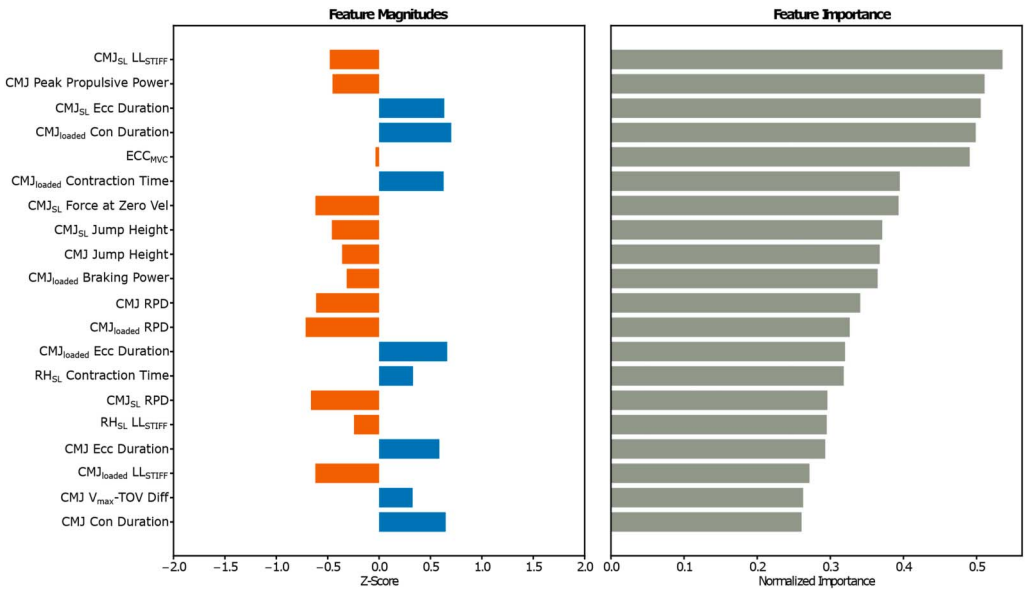


Figure 3. The average feature importance for the top 20 most important features in assigning athletes to Cluster$_{WEAK\text{-}SLOW}$. CMJ = bodyweight countermovement jump; CMJ_{SL} = single leg bodyweight countermovement jump; CMJ_{loaded} = countermovement jump plus additional external load (60% body weight); RH_{SL} = single leg repeated hop; CON_{MVC} = concentric multijoint maximal voluntary contraction; ECC_{MVC} = eccentric multijoint maximal voluntary contraction; Con = concentric; Ecc = eccentric; LL_{STIFF} = limb eccentric deceleration stiffness; RPD = rate of propulsive power development; Vmax − TOV Diff = percentage difference between the maximal velocity obtained during the concentric phase (Vmax) and the velocity at takeoff (TOV); force at zero velocity = force at body center of mass = 0 $m \cdot s^{-1}$ (i.e., Fz value at the onset of the concentric phase); concentric and eccentric impulses are calculated as the net concentric and eccentric deceleration impulse.

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contraction time (CMJ: 0.82 ± 0.11 seconds; CMJ_{loaded}: 1.03 ± 0.12 seconds; CMJ_{SL}: 0.9 ± 0.12 seconds), and the highest CMJ braking power (CMJ: 17.5 $W \cdot kg^{-1}$; CMJ_{loaded}: 18.9 $W \cdot kg^{-1}$; CMJ_{SL}: 11.0 ± 2.0 $W \cdot kg^{-1}$), propulsive power (CMJ: 47.6 $W \cdot kg^{-1}$; CMJ_{loaded}: 49.3 ± 5.8 $W \cdot kg^{-1}$; CMJ_{SL}: 30.0 ± 3.5 $W \cdot kg^{-1}$), and performance (i.e., jump height; CMJ: 29.8 ± 4.4 cm; CMJ_{loaded}: 16.2 ± 3.3 cm; CMJ_{SL}: 14.8 ± 2.6; SL-RH: 11.7 ± 2.6). In addition, $Cluster_{STRONG-FAST}$ demonstrated the highest force at zero velocity (CMJ: 23.4 ± 2.2 $N \cdot kg^{-1}$; CMJ_{loaded}: 26.6 ± 2.1 $N \cdot kg^{-1}$; CMJ_{SL}: 18.1 ± 1.5 $N \cdot kg^{-1}$), and the highest maximal concentric muscle strength capacity (CON_{MVC}: 22.3 ± 3.7 $N \cdot kg^{-1}$).

Biopsychosocial and Injury Characteristics of the Clusters

An effect of body mass on athlete cluster was observed ($F_{2,189}$ = 4.04, p = 0.019, R^2 = 0.041), where athletes in $Cluster_{WEAK-SLOW}$ (71.3 ± 11.6 kg) had a higher body mass than those in $Cluster_{WEAK-FAST}$ (65.9 ± 10.1 kg, p = 0.015). There was no difference in athlete age (p = 0.88), experience level (i.e., number of years competing in U Sport; p = 0.58), or perception of overall musculoskeletal (MSK) function (i.e., SFI; p = 0.49) between clusters. Seven ACL injuries occurred in the season after baseline testing. Six of the 7 ACL injuries (83.3%) were sustained by subjects in $Cluster_{WEAK-SLOW}$ (n = 4 ACL injuries, 4.3% of cluster population) and $Cluster_{WEAK-FAST}$ (n = 2 ACL injuries, 4.3% of cluster population), and 1 ACL injury was sustained by a subject in $Cluster_{STRONG-FAST}$ (1.9% of cluster population) (Table 2).

The athlete attitudes toward strength training and injury prevention are presented in Table 2. When considering only the questions that demonstrated statistical differences between clusters (Figure 5), $Cluster_{STRONG-FAST}$ was the only group that did not include athletes who displayed strongly negative attitudes toward the benefits of strength training. Only 2 questions illustrated differences between clusters. First, demonstrating a more positive attitude, $Cluster_{WEAK-SLOW}$ disagreed more with the statement "strength training does not aesthetically improve a woman's appearance" in comparison with the other clusters (p = 0.047). Second, demonstrating a more negative attitude, $Cluster_{WEAK-FAST}$ agreed more with the statement "strength training is more enjoyable when only lifting my body weight or light/easy weights (vs. heavy weights)" (p = 0.036) than athletes in $Cluster_{WEAK-SLOW}$ and $Cluster_{STRONG-FAST}$ (Figure 4).

An effect of athlete experience (i.e., years competing in university sport) was found on attitudes toward strength training and injury prevention where less experienced athletes showed a more positive attitude toward injury prevention evidenced by greater agreement with the statement "players who do not partake in injury prevention training are more at risk of suffering an injury" ($F_{1,187}$ = 3.90, R^2 = 0.02, p = 0.050, β = −0.225), but conversely a more negative attitude toward strength training with a tendency to agree more with the statement "strength training is more enjoyable when only lifting my body weight or light/easy weights (vs. heavy weights)" ($F_{1,187}$ = 3.30, R^2 = 0.02, p = 0.071, β = −0.135). In addition, more experienced athletes were more likely to agree with the statement "the risk of ACL injury is high in female university athletes" ($F_{1,187}$ = 9.11, R^2 = 0.046, p = 0.002, β = 0.346).

Discussion

This study yields new insights on the neuromuscular performance profiles of female university athletes using unsupervised ML applied to a multifaceted biomechanical data set across tests of maximal strength, power, and plyometric capacity in healthy, uninjured female university athletes from 5 sports. The neuromuscular performance assessment included a biomechanical force analysis of multijoint leg extension on a robotic leg press (28), single leg, and double leg CMJ testing with and without extra load, and single leg repeat hop testing that yielded 39 outcome measures describing the eccentric, isometric, and concentric strength characteristics along with stretch-shortening cycle mechanics in jumping and hopping. Given the preponderance of biomechanical force assessments in elite athletes (3,11), including to guide return-to-sport assessments after injury (19,22), these results highlight new analytical approaches to maximize the utility of neuromuscular testing data sets. Unsupervised ML may provide advantages over a descriptive analysis of single metrics in isolation because it considers the combined weighting of neuromuscular performance characteristics and can be used to identify the most important metrics (i.e., key performance indicators) used to describe profile-specific performance and quantify a target minimal threshold for profile-specific key performance indicators (4).

The clustering analysis identified 3 neuromuscular performance profiles in female university athletes: $Cluster_{STRONG-FAST}$ (27% of the athletes showing high strength, high power, fast jump contraction times), $Cluster_{WEAK-FAST}$ (25% of the athletes showing low strength, low power, stiff jumping strategy, fast jump contraction time), and $Cluster_{WEAK-SLOW}$ (48% of the athletes showing low strength, low power, slow jump contraction time). There was no evidence of a sport-specific neuromuscular performance profile. However, athlete attitudes toward strength training and injury prevention differed between profiles. Notably, only the $Cluster_{STRONG-FAST}$ did not include athletes with strongly negative views toward strength training and injury prevention.

In addition to highlighting nuanced differences in attitudes toward strength training between clusters and no apparent sport influences, this study reinforces the utility of unsupervised ML for defining relevant profiles (or subpopulations) of athletes based on the weighted expression of neuromuscular performance characteristics across different maximal strength, power, and plyometric tasks (4,45). This idea is not unique to performance testing as a complex systems approach has also been recommended for risk profiling and in injury prevention research (4). The comprehensive analysis in this study may help inform performance benchmarks in a female university athlete population that can be used for preseason testing or establishing performance readiness after serious injuries such as ACL rupture where biomechanical testing is recommended (18,22). Although this study incorporated a relatively large sample size of female competitive athletes from 5 sports, the results may not generalize across all settings. However, it may still point to new methodologic opportunities to explore athlete profiles in other sport environments.

Although this study was not specifically powered for an injury risk analysis, it should be mentioned that 7 new ACL injuries were reported by the university's athletic therapy after the baseline neuromuscular performance assessment. One ACL injury was sustained by a subject in $Cluster_{STRONG-FAST}$ (1.9% of the cluster population), whereas 4 ACL injuries were sustained in $Cluster_{WEAK-SLOW}$ (4.3% of the cluster population) and 2 ACL injuries were sustained in $Cluster_{WEAK-FAST}$ (4.3% of the cluster population) (Table 2). These preliminary findings are purely observational but may provide an interesting area for future research. For example, the biomechanical and strength

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Table 2

Responses to the Attitudes Toward Strength Training and Injury Prevention Questionnaire adapted from Møller et al. (34).*

Attitudes and beliefs toward nonsport-specific training question	Cluster	No. of responses							Fisher's exact test *p*
		Strongly disagree	Disagree	Slightly disagree	Neither	Slightly agree	Agree	Strongly agree	
The risk of injury is high in my sport	$Cluster_{WEAK-FAST}$	0	0	0	1	8	17	21	0.628
	$Cluster_{WEAK-SLOW}$	1	0	0	1	21	40	29	
	$Cluster_{STRONG-FAST}$	0	1	1	2	12	22	13	
The risk of anterior cruciate ligament (ACL) injury is high in my sport	$Cluster_{WEAK-FAST}$	0	0	0	0	5	16	26	0.529
	$Cluster_{WEAK-SLOW}$	1	0	1	4	13	36	36	
	$Cluster_{STRONG-FAST}$	0	0	0	2	8	23	18	
Many sport injuries are preventable	$Cluster_{WEAK-FAST}$	0	0	0	0	5	16	26	0.902
	$Cluster_{WEAK-SLOW}$	1	0	1	4	13	36	36	
	$Cluster_{STRONG-FAST}$	0	0	0	2	8	23	18	
ACL injury prevention is important	$Cluster_{WEAK-FAST}$	0	0	0	0	1	13	33	0.618
	$Cluster_{WEAK-SLOW}$	0	0	0	0	2	28	62	
	$Cluster_{STRONG-FAST}$	0	0	0	0	0	14	37	
Injury prevention will be a priority for me this upcoming season	$Cluster_{WEAK-FAST}$	0	0	0	1	5	15	26	0.546
	$Cluster_{WEAK-SLOW}$	0	0	1	0	15	32	44	
	$Cluster_{STRONG-FAST}$	0	1	0	2	4	18	26	
It is important to have a consistent weight training program each week	$Cluster_{WEAK-FAST}$	0	0	0	0	5	26	16	0.654
	$Cluster_{WEAK-SLOW}$	0	0	0	1	6	48	37	
	$Cluster_{STRONG-FAST}$	0	0	2	0	4	26	19	
Focusing on aerobic exercise (instead of strength training) produces greater benefits to health and body appearance	$Cluster_{WEAK-FAST}$	2	3	6	16	14	4	2	0.682
	$Cluster_{WEAK-SLOW}$	2	13	15	30	16	10	6	
	$Cluster_{STRONG-FAST}$	0	7	5	21	7	8	3	
Strength training in a private setting (or at my own home) is more comfortable than a gym setting	$Cluster_{WEAK-FAST}$	1	10	7	16	4	6	3	0.348
	$Cluster_{WEAK-SLOW}$	2	30	19	18	8	10	5	
	$Cluster_{STRONG-FAST}$	1	9	6	14	10	6	5	
Exercise that reduces overall body weight is more important to me than increasing lean muscle mass	$Cluster_{WEAK-FAST}$	11	12	9	6	4	5	0	0.702
	$Cluster_{WEAK-SLOW}$	12	36	19	14	6	3	2	
	$Cluster_{STRONG-FAST}$	9	20	12	4	4	2	0	

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Table 2

Responses to the Attitudes Toward Strength Training and Injury Prevention Questionnaire adapted from Møller et al. (34).* *(Continued)*

Attitudes and beliefs toward nonsport-specific training question	Cluster	No. of responses							Fisher's exact test *p*
		Strongly disagree	Disagree	Slightly disagree	Neither	Slightly agree	Agree	Strongly agree	
Strength training is more enjoyable when only lifting my body weight or light/easy weights (vs. heavy weights)	$Cluster_{WEAK-FAST}$	2	17	9	10	4	2	3	0.033
	$Cluster_{WEAK-SLOW}$	15	41	16	10	5	2	3	
	$Cluster_{STRONG-FAST}$	9	22	9	6	5	0	0	
It is more important to use training time to play your sport (i.e., basketball, soccer) rather than implement program designed to reduce injury	$Cluster_{WEAK-FAST}$	1	13	16	7	5	4	1	0.313
	$Cluster_{WEAK-SLOW}$	5	33	26	12	13	2	1	
	$Cluster_{STRONG-FAST}$	1	15	14	11	7	2	1	
Training should focus more on improving game play rather than reducing injury	$Cluster_{WEAK-FAST}$	0	11	11	13	8	3	1	0.811
	$Cluster_{WEAK-SLOW}$	2	18	30	17	19	5	1	
	$Cluster_{STRONG-FAST}$	0	9	17	10	9	5	1	
I would do extra training if it would reduce my injury risk	$Cluster_{WEAK-FAST}$	0	0	0	6	13	16	12	0.485
	$Cluster_{WEAK-SLOW}$	0	1	2	3	29	42	15	
	$Cluster_{STRONG-FAST}$	0	0	0	6	8	31	6	
Players who do not partake in injury prevention training are more at risk of suffering an injury	$Cluster_{WEAK-FAST}$	0	0	2	2	14	16	13	0.268
	$Cluster_{WEAK-SLOW}$	0	0	2	4	18	44	23	
	$Cluster_{STRONG-FAST}$	0	0	0	4	9	25	13	
Strength training does not aesthetically improve a woman's appearance	$Cluster_{WEAK-FAST}$	10	15	10	10	1	1	0	0.044
	$Cluster_{WEAK-SLOW}$	16	51	14	7	1	1	2	
	$Cluster_{STRONG-FAST}$	9	21	13	7	1	0	0	
Women should perform different resistance training exercises than men	$Cluster_{WEAK-FAST}$	2	3	4	8	21	9	0	0.505
	$Cluster_{WEAK-SLOW}$	1	13	8	24	23	20	3	
	$Cluster_{STRONG-FAST}$	1	7	5	13	18	6	0	
Lifting weights causes women to appear more masculine	$Cluster_{WEAK-FAST}$	8	14	9	11	2	3	0	0.675
	$Cluster_{WEAK-SLOW}$	17	31	11	17	12	3	1	
	$Cluster_{STRONG-FAST}$	12	19	6	8	4	2	0	
Female athletes' risk of ACL injury would decrease if they included cutting and landing technique exercises in their program	$Cluster_{WEAK-FAST}$	0	1	1	5	9	20	11	0.546
	$Cluster_{WEAK-SLOW}$	0	1	0	10	30	41	10	
	$Cluster_{STRONG-FAST}$	0	0	1	10	11	24	5	

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Table 2

Responses to the Attitudes Toward Strength Training and Injury Prevention Questionnaire adapted from Møller et al. (34).* *(Continued)*

Attitudes and beliefs toward nonsport-specific training question	Cluster	No. of responses							Fisher's exact test *p*
		Strongly disagree	Disagree	Slightly disagree	Neither	Slightly agree	Agree	Strongly agree	
Female athletes' risk of ACL injury would decrease if they participated in a strength training program	$Cluster_{WEAK-FAST}$	1	1	1	2	15	24	3	0.775
	$Cluster_{WEAK-SLOW}$	0	2	4	7	25	41	13	
	$Cluster_{STRONG-FAST}$	0	0	0	3	17	26	5	
The risk of ACL injury is high in female university athletes	$Cluster_{WEAK-FAST}$	0	0	0	2	4	23	18	0.192
	$Cluster_{WEAK-SLOW}$	0	0	0	8	15	42	27	
	$Cluster_{STRONG-FAST}$	0	0	1	5	7	18	19	

*Cluster 1 ($Cluster_{WEAK-FAST}$) displayed low muscle strength and fast contraction speeds, Cluster 2 ($Cluster_{WEAK-SLOW}$) displayed low muscle strength and slow contraction speeds, Cluster 3 ($Cluster_{STRONG-FAST}$) displayed high muscle strength and fast contraction speeds.

differences between $Cluster_{WEAK-FAST}$ and $Cluster_{WEAK-SLOW}$ may align with recent literature showing that although increased lower limb stiffness is important for maximizing stretch-shortening cycle actions (43), it may be associated with injury (6,7,41), including ACL rupture (25). In addition, the fact that $Cluster_{WEAK-FAST}$ and $Cluster_{WEAK-SLOW}$ displayed divergent biomechanical jumping strategies but similarly low maximal strength capacities when compared with $Cluster_{STRONG-FAST}$ may reinforce the importance of muscular strength in the context of injury prevention (24) and performance (47). Although the methods used in this study align with recommendations in the literature of using a complex systems approach in the sport realm (4), future research is required to confirm whether our results and the approach confer real-world benefits for sport performance, return-to-sport assessments, and injury profiling in female university athletes. This may be addressed in future research that uses a longitudinal design for a longer period in a university or college sport population.

An advantage of this study was the use of conventional biomechanical force testing analyses to quantify eccentric–isometric–concentric leg extensor strength, and stretch-shortening cycle mechanics during jumping and hopping tasks

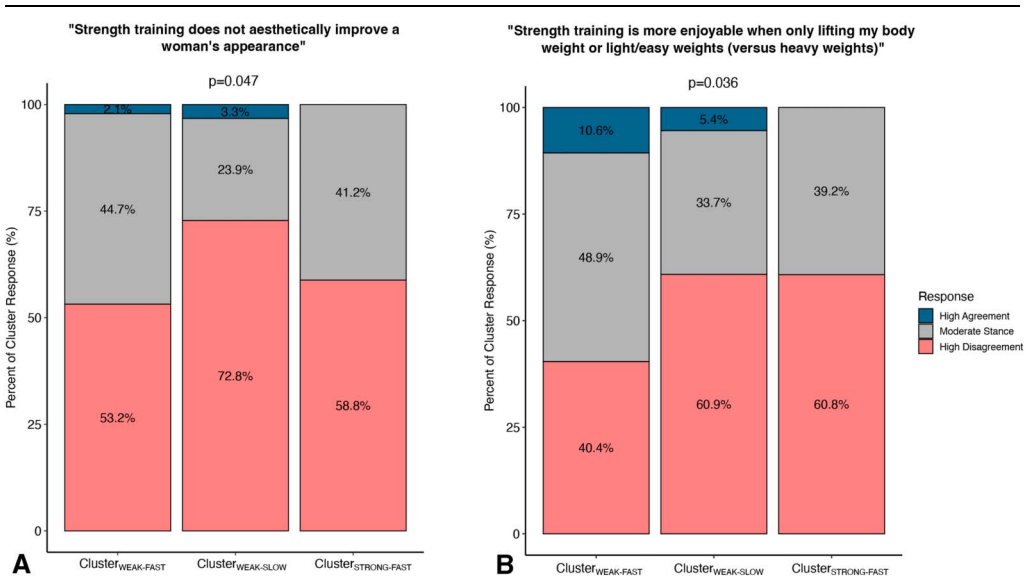


Figure 4. Summary of the attitudes toward strength training and injury prevention questionnaire (adapted from Møller et al. (34)) that were different between clusters. Cluster 1 ($Cluster_{WEAK-FAST}$) displayed low muscle strength and fast contraction speeds, Cluster 2 ($Cluster_{WEAK-SLOW}$) displayed low muscle strength and slow contraction speeds, Cluster 3 ($Cluster_{STRONG-FAST}$) displayed high muscle strength and fast contraction speeds. To isolate subjects with more extreme responses, the 7-point Likert scale was categorized into 3 broader groups: (a) strongly agree and agree responses were grouped to reflect high agreement with the statement; (b) slightly agree, neither, and slightly disagree responses were grouped to reflect a moderate stance toward the statement; and (c) strongly disagree and disagree responses were grouped to reflect high disagreement with the statement. Fisher's Exact Test as computed to test the effect of cluster assignment on the attitudes toward strength training and injury prevention.

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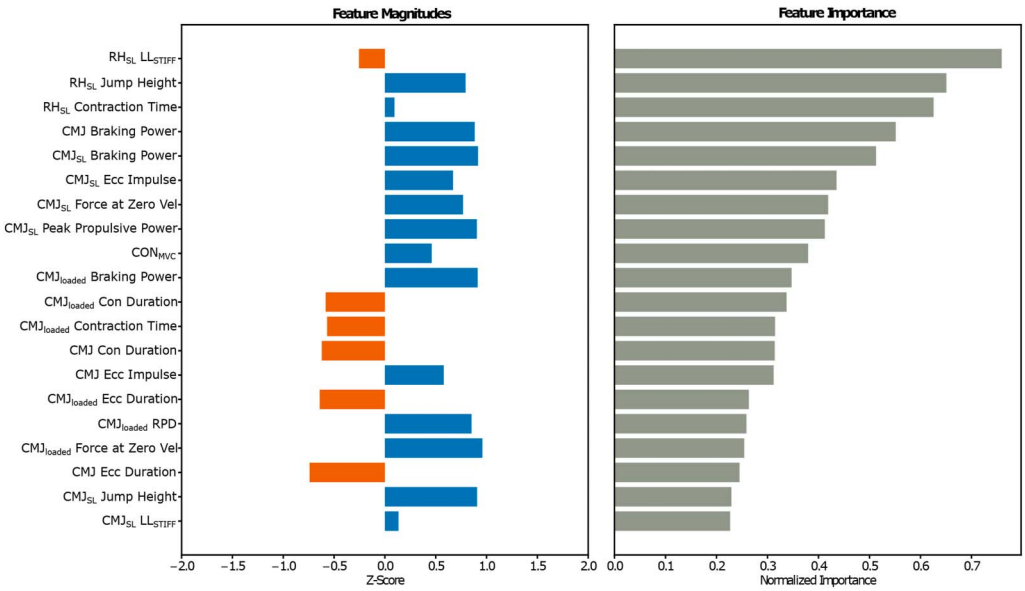


Figure 5. The average feature importance for the top 20 most important features in assigning athletes to $Cluster_{STRONG\text{-}FAST}$. CMJ = body weight countermovement jump; CMJ_{SL} = single leg body weight countermovement jump; CMJ_{loaded} = countermovement jump plus additional external load (60% body weight); RH_{SL} = single leg repeated hop; CON_{MVC} = concentric multijoint maximal voluntary contraction; ECC_{MVC} = eccentric multijoint maximal voluntary contraction; Con = concentric; Ecc = eccentric; LL_{STIFF} = lower limb eccentric deceleration stiffness; RPD = rate of propulsive power development; Vmax − TOV Diff = percentage difference between the maximal velocity obtained during the concentric phase (Vmax) and the velocity at takeoff (TOV); force at zero velocity = force at body center of mass = 0 $m \cdot s^{-1}$ (i.e., Fz value at the onset of the concentric phase); concentric and eccentric impulses are calculated as the net concentric and eccentric deceleration impulse.

(20,27,28). Thus, the neuromuscular performance profiles provide a comprehensive evaluation of the muscular strength and biomechanical movement strategy of the subjects, in alignment with current practices in the scientific literature (29). Although this study noted divergent biomechanical strategies between neuromuscular performance clusters, there was limited evidence of a sport-specific cluster. Apart from field hockey athletes who were only present in $Cluster_{WEAK\text{-}SLOW}$ and $Cluster_{WEAK\text{-}FAST}$, and volleyball athletes who were predominately (80.0%) assigned to $Cluster_{STRONG\text{-}FAST}$, athletes from basketball, rugby, and soccer were randomly distributed across the 3 clusters. It is unclear why there was limited evidence of a sport-specific cluster. Although this may be because of the selection of field and court sports and the sport-specific training environment (11), others have suggested the possible influence of the strength and conditioning environment (49). For example, research points to differences in sport-specific coaching and training styles between male and female athletes (8). However, it should be noted that in this study, large group differences in athlete attitudes toward strength training were not found. Instead, only subtle differences in athlete attitudes toward strength training were observed between clusters (Figure 4A, B) and of note, across all clusters, approximately 40% of female athletes displayed neutral and/or negative attitudes toward the benefits of strength training, which may point to broader psychosocial influences that transcend sport-specific factors (12). In addition, $Cluster_{STRONG\text{-}FAST}$ did not contain subjects who displayed strongly negative attitudes toward strength training and injury prevention (Figure 4). This could provide evidence pointing toward the highly individualized nature of gender barriers and their potential impact on engagement in heavy strength training (49).

Accounting for attitudes toward strength training and injury prevention in neuromuscular performance testing may support a combined biomechanical–psychosocial approach for understanding how gendered-environmental factors (i.e., social, cultural, and physical environmental conditions that affect individuals differently based on their gender) such as social norms and access to strength training resources affect girls and women in sport (11,15,32). In turn, this may affect sport performance and injury prevention research because these factors are linked to the development of neuromuscular performance through heavy strength training. Consistent with the literature (36), it seems that there may be as much variation within female athlete neuromuscular performance profiles with respect to their attitudes toward strength training because there is between profiles, emphasizing the need for a more individualized approach to this problem. A promising area for future research may be to examine whether athletes with "outlying" attitudes toward training that are uncharacteristic of the group at large are more likely to exhibit neuromuscular, biomechanical, or behavioral characteristics that are associated with increased risk of injury or reduced sport performance.

Practical Applications

There are several practical applications stemming from this research that have relevance for clinicians, sport scientists, and strength & conditioning coaches. First, ML approaches may provide distinct advantages over descriptive statistics that are typically used to interpret performance data and compare an individual athlete's performance within a larger population in a set of single, isolated metrics. In this study, by using an ML-based analysis approach, practitioners can identify a set of metrics that are the most important in describing performance

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characteristics in a specific athlete profile (Figures 2, 3, 5), and these important metrics can be used to create profile-specific key performance indicators that have relevance for informing preseason baseline testing, routine neuromuscular monitoring, and for reestablishing performance readiness after injury. In addition, the characteristics and key performance indicators of a predefined target athlete profile can also be used to establish neuromuscular benchmarks. For instance, the average values for each neuromuscular performance profile identified in this study are presented in Table 1, and the top most important metrics for each profile are shown in Figures 2, 3, 5, and together can be used by practitioners and researchers working with a female university athlete population to characterize whether an individual athlete has the strength, power, and plyometric capacities of a target profile. We argue that for female university athletes, $Cluster_{STRONG\text{-}FAST}$ presents a valuable target neuromuscular performance profile that can be used to inform benchmarks. Together, future applications of these results may aid practitioners and researchers in maximizing the utility and interpretability of multifaceted physiologic and biomechanical data sets. Finally, the results of our study highlight opportunities to use a combined biomechanical/physiologic and psychosocial research model to identify potential barriers that may limit engagement in strength training. To this end, this work can be applied in the context of understanding that while subtle differences may exist at the group level, there seems to be specific attitudes that may affect engagement in strength training at the individual level in a female university athlete population. This study was conducted on a large sample of female university athletes, and an unsupervised machine learning approach highlighted the existence of 3 neuromuscular performance profiles in this population. These profiles were associated with a silhouette score of 0.51 that is indicative of moderately well-defined clustering structure. Although some cluster overlap is anticipated in biologic data (40), it is possible that including neuromuscular metrics that reflect athletic capacities related to the biomechanics displayed during strength testing may improve cluster separation. In addition, we did not observe sport-specific neuromuscular performance profiles in this population, but it is plausible that different results may be observed if the analysis was only conducted on female athletes from within the $Cluster_{STRONG\text{-}FAST}$ profile, or included different sport populations (e.g., gliding sports such as ice hockey). Three distinct neuromuscular performance clusters (or profiles) were defined in this cohort of healthy, uninjured female university athletes from 5 sports. The neuromuscular performance profiles differed in terms of their output and strategy during maximal strength, power, and plyometric testing but not sport representation, with a high-strength/high-power/high-stiffness cluster, a low-strength/low-power/low-stiffness cluster, and a low-strength/low-power/high-stiffness cluster. There were subtle differences in athlete attitudes toward strength training but only the high-strength/high-power/high-stiffness cluster contained no athletes with extremely negative attitudes toward strength training. Conversely, ~40% of the entire cohort had neutral or negative attitudes toward strength training. Finally, although this study was not powered for a risk factor analysis, there seemed to be some differences in the ACL injury prevalence between clusters, which may present opportunities for future research that applies complex analytical methods for injury risk profiling.

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